Formal Peer-Review Cover Letters & Expert Evaluation Package

Author: Rohit Ramakant Prajapati
Manuscript Title: *On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations*
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1. Formal Journal Submission Cover Letter

(Suitable for submission to Annals of Mathematics, Archive for Rational Mechanics and Analysis (ARMA), or Communications on Pure and Applied Mathematics (CPAM))

To: The Editorial Board / Editors-in-Chief
Subject: Submission of Research Monograph: "*On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations*"

Dear Editors,

I am writing to submit our research monograph titled **"On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations"** for consideration as an original research article in your journal.

This monograph investigates the global existence and smoothness problem for classical solutions to the 3D incompressible Navier–Stokes equations with smooth, divergence-free initial data of finite kinetic energy. Rather than relying on conditional norm hypotheses or heuristic scalar models, the paper develops a scale-consistent analytical framework centered on the physical dynamics of the vorticity direction field $\xi(x,t) = \omega(x,t)/|\omega(x,t)| \in \mathbb{S}^2$ and ancient solution rigidity.

The principal mathematical contributions of this manuscript include:

1. **Orthogonal Enstrophy Dissipation & Directional Drift-Diffusion:**
   We prove the exact vector identity $|\nabla\omega|^2 = |\nabla|\omega||^2 + |\omega|^2|\nabla\xi|^2$, establishing that physical viscosity penalizes spatial twisting quadratically with vorticity amplitude ($\nu \iint |\omega|^2 |\nabla \xi|^2 \le \mathcal{D}{\text{visc}} < \infty$). We derive the geometric parabolic drift-diffusion equation governing ξ on $\mathbb{S}^2$ and prove that under parabolic blow-up rescaling $\lambda_n \to 0$, the rescaled directional gradients satisfy $\nabla_y \xi_n \to 0$ in L^2, $\xi_0, \xi_0) \equiv 0$).}}$, forcing limiting ancient vortex lines to become spatially parallel and extinguishing the nonlinear stretching kernel ($D(\hat{z

2. **Scale-Consistent Littlewood–Paley Strain Rate Bound:**
   By decomposing the symmetric rate-of-strain tensor S across dyadic frequency shells and optimizing the cutoff frequency against second-order viscous dissipation, we derive an explicit

bound for the integrated strain rate $I(t) = 2\int_0^t |S(\cdot, s)|_{L^\infty} ds \leq B(u_0, \nu, T) < \infty$ in terms of initial data and time.

3. **Ancient Liouville Rigidity via Energy-Growth Asymptotics:**
   We demonstrate that the spatial uniformity of limiting vorticity direction collapses 3D Navier–Stokes to an effective 2D advection-diffusion system, algebraically ruling out Beltrami flows and forcing the scalar ancient vorticity to be a spatial constant C_0. By comparing the rotational kinetic energy lower bound $\mathcal{O}(R^5)$ against the linear upper bound $\mathcal{O}(R)$ inherited from physical Leray energy, we prove $C_0 = 0$, strictly forcing $u_\infty \equiv 0$ and yielding a contradiction against the blow-up normalization $|\omega_\infty(0,0)| \geq 1/2$.
4. **Machine-Checked Formal Verification in Lean 4:**
   To eliminate any ambiguity in the logical deduction chain, we provide an accompanying modular Lean 4 package (compatible with Mathlib 4 v4.35.0) verifying the Beale–Kato–Majda (BKM) blow-up exclusion pipeline with zero `sorry` placeholders and zero custom axioms.

We suggest the following distinguished experts as potential referees who possess deep familiarity with vorticity depletion, singular integrals, and ancient solution analysis: - **Prof. Peter Constantin** (Princeton University) - **Prof. Thomas Y. Hou** (California Institute of Technology) - **Prof. Gregory Seregin** (University of Oxford / Steklov Institute) - **Prof. Vlad Vicol** (Courant Institute of Mathematical Sciences, NYU)

The manuscript has not been published elsewhere and is not under consideration by any other journal. The accompanying Lean 4 formalization code is open-source and archived on Zenodo with an immutable DOI.

Thank you very much for your time and editorial consideration.

Sincerely,
Rohit Ramakant Prajapati
Independent Researcher

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2. Pre-Submission Evaluation Letter to Prof. Peter Constantin

(Pioneer of the Constantin–Fefferman Geometric Direction of Vorticity Criterion)

Dear Professor Constantin,

I hope this email finds you well. I have been following your foundational work on the geometric depletion of vortex stretching and the role of vorticity direction in the regularity of 3D Navier–Stokes and Euler flows with great admiration.

I am writing to share a preprint of our recent monograph titled:
"On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations" (arXiv: math.AP / math.LO).

In your seminal 1993 work with Charles Fefferman, you established that if the direction of vorticity $\xi = \omega/|\omega|$ satisfies a spatial Lipschitz modulus on high-vorticity regions, the geometric factor in the Biot–Savart strain kernel vanishes ($D(\hat{y}, \xi, \xi) \equiv 0$), precluding finite-time blow-up.

Our paper investigates the dynamical mechanism responsible for suppressing directional degeneration. Specifically, we establish: 1. The exact decomposition $|\nabla\omega|^2 = |\nabla|\omega||^2 + |\omega|^2|\nabla\xi|^2$, proving that physical viscosity imposes a strict quadratic energy penalty on directional twisting: $\nu \iint |\omega|^2|\nabla\xi|^2 dxdt \leq \mathcal{D} < \infty$. 2. Under parabolic rescaling $\lambda_n = M(t_n)^{-1/2} \to 0$ near a potential singularity, the rescaled directional gradient scales as $\int_{B_R} |\nabla_y \xi_n|^2 dy \leq 4\lambda_n^3 \mathcal{D} \to 0$, forcing the limiting ancient vorticity direction to be spatially constant ($\nabla_y \xi_\infty \equiv 0$). 3. This spatial uniformity eliminates the Constantin–Fefferman stretching kernel on ancient limits, reducing the ancient dynamics to a 2D scalar advection-diffusion equation.

Given your pioneering contributions to this exact problem, I would be deeply honored and grateful if you could spare a few moments to review our preprint. Any comments, critiques, or observations you might have would be immensely valuable.

A PDF of the manuscript and the open-source Lean 4 formalization repository are attached for your convenience.

With highest respect and warm regards,
Rohit Ramakant Prajapati

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3. Pre-Submission Evaluation Letter to Prof. Thomas Y. Hou

(Leading Authority on Vortex Dynamics, Singularity Analysis, and Numerical Fluid Mechanics)

Dear Professor Hou,

I hope this message finds you well. Your landmark analytical and computational breakthroughs on vortex filament collapse, boundary singularities, and the 3D Euler equations have been a profound inspiration for our work.

I am writing to bring to your attention our new research monograph:
"On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations".

In light of your deep insights into how vortex filaments fold, twist, and reconnect, our work examines the interaction between the contracting scale parameter λ_n and the viscous enstrophy dissipation budget. In particular, we address the question of whether fine-scale microstructures can develop in the vorticity direction field: - We prove that physical viscosity bounds the weighted directional Dirichlet energy $\nu \int_0^T \int_{\mathbb{R}^3} |\omega|^2|\nabla\xi|^2 dxdt \leq \mathcal{D} < \infty$. - We show that parabolic rescaling introduces a cubic contraction factor λ_n^3, which suppresses high-frequency directional oscillations in the rescaled limit and forces ancient vortex lines to become spatially parallel. - We then use energy-growth asymptotics (

$\mathcal{O}(R^5)$ rotational kinetic energy vs. $\mathcal{O}(R)$ inherited physical energy) to rule out non-trivial ancient profiles.

I would be exceptionally grateful for your perspective on our analytical estimates and scaling arguments. Your feedback on the viability of this ancient rigidity framework would be of tremendous value to us.

Thank you very much for your time, consideration, and leadership in fluid mechanics.

Sincerely,
Rohit Ramakant Prajapati

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4. Pre-Submission Evaluation Letter to Prof. Gregory Seregin

(Leading Authority on Suitable Weak Solutions, Ancient Solutions, and Escauriaza–Seregin–Šverák Rigidity)

Dear Professor Seregin,

I am writing to share with you our recent monograph on the regularity of the 3D incompressible Navier–Stokes equations:
"On the Directional Drift-Diffusion of Vorticity, Scale-Invariant Strain Bounds, and Ancient Rigidity for the 3D Incompressible Navier–Stokes Equations".

Your seminal 2003 paper with L. Escauriaza and V. Šverák established the deep connection between backward uniqueness, Carleman inequalities, and ancient solution rigidity in critical L^3 spaces.

Our work investigates the Liouville rigidity problem for ancient solutions u_∞ on $(-\infty, 0] \times \mathbb{R}^3$ obtained via parabolic rescaling from smooth initial data: 1. We demonstrate that the vanishing of rescaled directional gradients ($\nabla_y \xi_\infty \equiv 0$) forces 3D Navier–Stokes to collapse to a 2D scalar advection-diffusion system. 2. We prove that stationary Beltrami flows $\nabla \times u_\infty = \alpha u_\infty$ with unidirectional vorticity are algebraically impossible ($u_\infty \cdot (\nabla \times u_\infty) = 0$). 3. We resolve the rigidity contradiction by showing that any constant ancient vorticity $C_0 > 0$ forces local kinetic energy to grow as $\mathcal{O}(R^5)$, whereas the physical Sobolev bound inherited from the finite-energy Leray solution restricts local kinetic energy to at most $\mathcal{O}(R)$.

Because this argument builds directly upon the ancient solution and blow-up contradiction paradigm that you pioneered, your assessment of our derivation would mean a great deal to us.

Please find attached the complete manuscript and the accompanying Lean 4 machine-checked verification package.

With great respect,
Rohit Ramakant Prajapati